
Calibration data
residuals and contexts
Weighted risk map
$(g, \lambda) \mapsto \bar{r}_g(\lambda)$
Budget-aware radius choice
pick (α_t, λ_t) so that
$\bar{r}_{g_t}(\lambda_t) \leq \alpha_t$
Tube MPC
solve tightened problem
u_t
Plant and learned model
state x_t, input
u_t, error e_t
valid refresh requires
sample splitting/conf. spending
On the calibration event Ω, hard-feasible implemented
plans with $\sum_t p_t \leq \varepsilon$ satisfy $\mathbb{E}[N_T \mid \Omega] \leq \varepsilon$.